



LUND
UNIVERSITY

Centre for Mathematical Sciences
Mathematics, Faculty of Science

Written Examination
Group and Ring Theory
MATP33
15 August 2023
Duration: 08:00–13:00

Answers

1. (a) $G = \mathbb{Z}$.
6. $(\mathbb{Z}/3\mathbb{Z})^6$, so the elementary divisors are 3, 3, 3, 3, 3, 3, and these are also the invariant factors.